When Short Responses Are Hard to Distinguish in Translation Audits

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Introduction

Workplace surveys often contain **brief or repeated answers**. Translation makes these responses accessible to English-centered analysis tools, including resources used in mental-health research.

We examine what reference-free audit tools reveal about Portuguese-to-English survey translations and how **response length, repetition and lexical coverage** shape their interpretation.

Data and methods

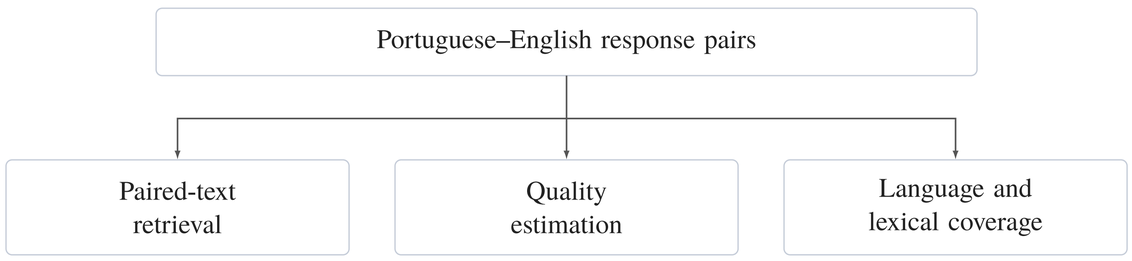
The corpus contains **44,118 response pairs** from **14,706 survey-specific records** across four Brazilian workplace survey cohorts from 2023 to 2025. Each record contributes three answers. One third of pairs come from mental-health or well-being prompts.

For **paired-text retrieval**, Qwen3-Embedding-4B [1] ranks all 44,118 candidates in the other language. The paired target or an **exact normalized duplicate** counts as a correct match. We also tested the 8B encoder.

We estimate **translation quality** with COMETKiwi DA-XL [2], using the Portuguese source and English translation.

We use Lingua for **language flags** and EmoAtlas [3] to measure **emotion-lexicon coverage** in each language.

Retrieval, quality estimation and coverage are **interpreted separately**.



Interpretation

Retrieval measures **how distinguishable a paired text is** within this corpus. A missed match can reflect brief or repeated wording.

COMETKiwi follows a different pattern across response lengths. Reporting the measures separately keeps their meanings clear. **Bilingual human ratings** would calibrate the quality estimates.

Limitations

Results describe **one language pair** and one workplace-survey setting. Translation provenance is incomplete, and encoder comparisons remain within the Qwen3 family.

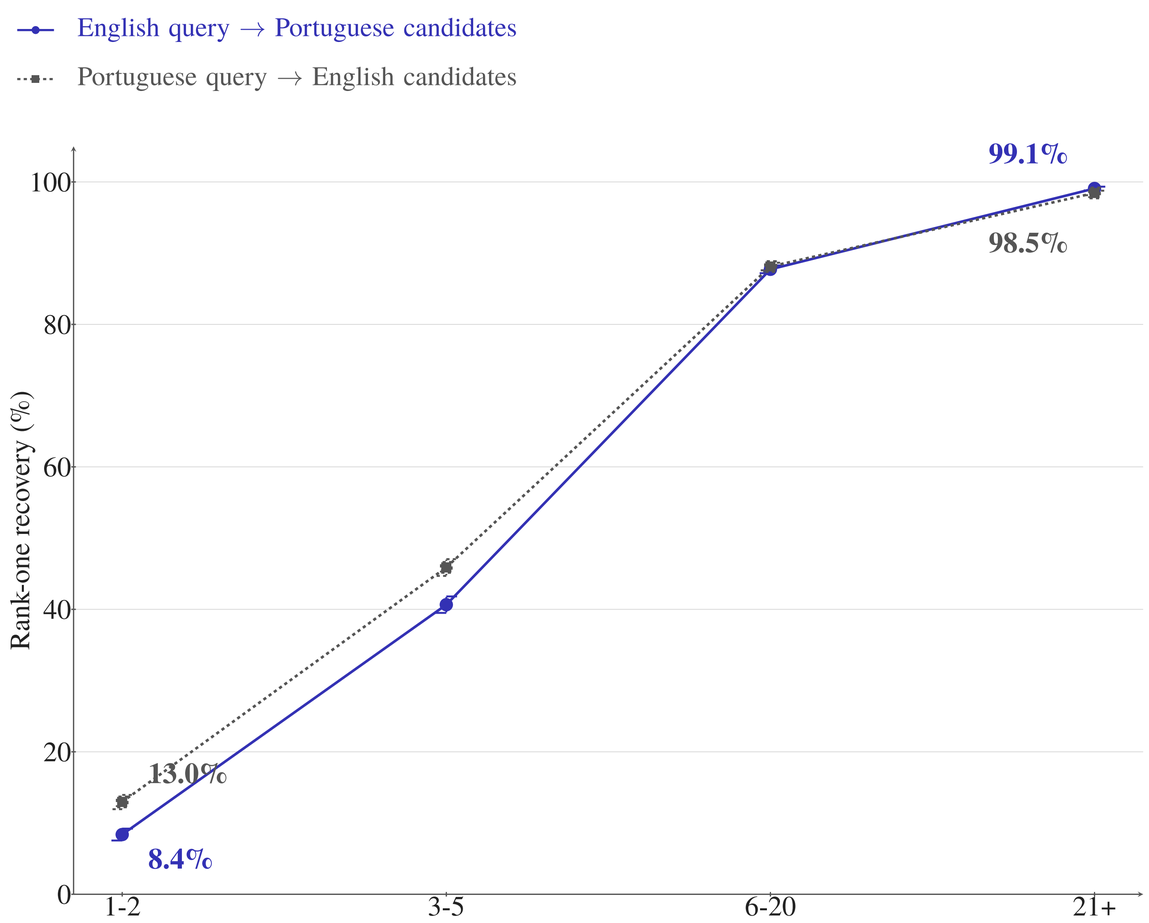
Assessing translation accuracy and emotional or cultural equivalence requires **human assessment and validation** in the target domain.

Survey comments remain **confidential**. The public code includes frozen configurations and synthetic tests.

Results

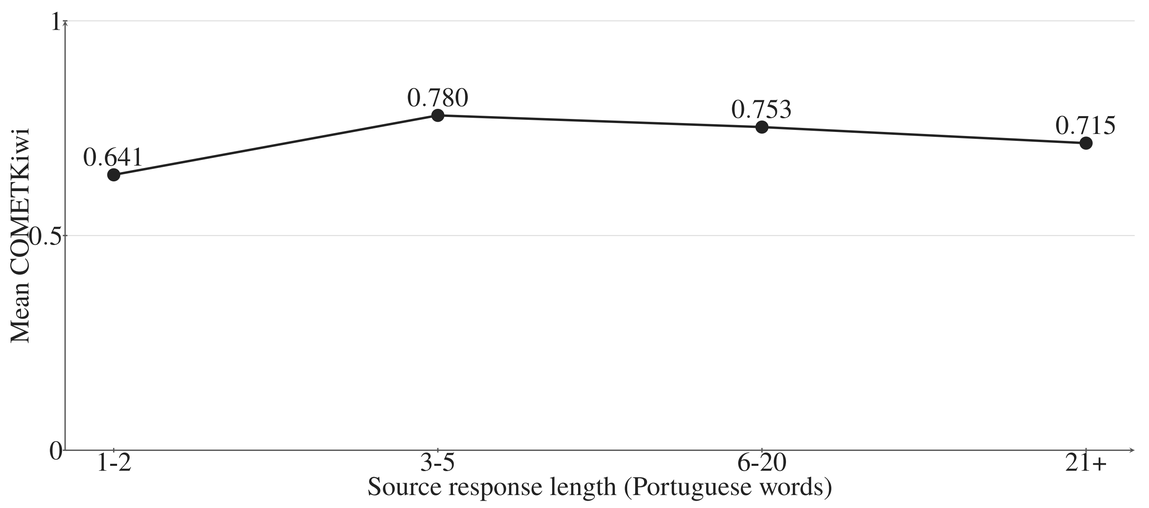
Retrieval by response length

Rank-one recovery finds the paired target or an exact normalized duplicate in the first position.



Error bars show **95% bootstrap intervals** from 2,000 resamples, clustered by survey-specific record. Every query searches the full set of 44,118 candidates.

Quality estimates differ from retrieval



COMETKiwi scores are **model estimates**. Calibration against bilingual ratings is needed for this domain.

Responses with zero emotion-lexicon coverage

Portuguese	24.67%
English	23.48%

Zero coverage limits lexical comparisons across languages.

Selected references

[1] Zhang et al. (2025). Qwen3 Embedding. arXiv:2506.05176.
[2] Rei et al. (2023). Scaling up CometKiwi. WMT 2023.
[3] Semeraro et al. (2025). EmoAtlas. Behavior Research Methods.

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Code